Affine-invariant ensemble HMC (PEACHES, PEANUTS): benchmark study *(corrected dual averaging)*

Abstract

*Affine-invariant ensemble HMC (*PEACHES, PEANUTS*) inherits the optimal $d^{-1/4}$ step-size scaling of tuned Hamiltonian methods, matches well-tuned NUTS in per-gradient efficiency, and is far more robust: it learns its preconditioner from the live ensemble, needs no hand-tuned mass matrix, and does not collapse when the warmup budget is short. This version corrects a dual-averaging bug found while validating the scaling: a spurious $1/(m+t_0)$ factor in the step-size gain made the acceptance target effectively open-loop, so the adapter never honoured* `target_accept` *and the step size did not scale with dimension. The fix (one line, all samplers) is an* efficiency *correction, not a correctness one — prior samples were unbiased — and with it the $d^{-1/4}$ law is reproduced by the sampler's own adaptation.*

We benchmark the two new algorithms (PEACHES: ensemble ChEES-HMC; PEANUTS: ensemble NUTS) against the affine but gradient-free stretch move, a non-affine gradient ensemble (CHESS), and well-tuned NumPyro NUTS. All numbers are 3-seed medians (synthetic).

Metrics (how everything is measured). Let $\{x_{c,i}^{(t)}\}$ be the post-warmup draws (t=iteration, c=chain/walker, i=coordinate), with closed-form truth (μ_i, σ_i) from the target.

- **Bias** (no reference run needed): for each coordinate the estimator is the pooled sample mean $\hat{\mu}_i = \mathrm{mean}_{t,c}\, x_{c,i}$; we report $\max_i |\hat{\mu}_i - \mu_i|/\sigma_i$, the worst-coordinate mean error in units of the true standard deviation. We also report the *z-score* $|\hat{\mu}_i - \mu_i|/\mathrm{MCSE}_i$, where MCSE is the ArviZ Monte-Carlo standard error; $z \approx 1$–2 means the estimate is statistically consistent with truth (an honest error bar), not residual bias.
- **IAT** (τ): integrated autocorrelation time $= N_{\mathrm{draws}}/\mathrm{ESS}$ with ArviZ rank-normalized split-chain ESS; $\tau < 1$ signals antithetic (negatively correlated) draws.
- **ESS/grad**: $\mathrm{ESS}_{\min}$ divided by the *total* gradient evaluations (production only, counted exactly per sampler). For an HMC method this equals $1/(\tau \bar{L})$ with $\bar{L}$ the mean leapfrog steps per trajectory — the natural cross-sampler efficiency (§2).
- **Convergence**: split-$\hat{R}$ and split-half ESS ratio; we treat each walker as a chain. All quantities use float64.

1 A dual-averaging bug, and why it was invisible

The Nesterov dual-averaging step-size update (Hoffman & Gelman 2014) is

$$\bar{H}_m = \left(1 - \tfrac{1}{m+t_0}\right)\bar{H}_{m-1} + \tfrac{1}{m+t_0}\,(\delta - \alpha_m), \qquad \log \epsilon_m = \mu - \frac{\sqrt{m}}{\gamma}\,\bar{H}_m.$$

The $1/(m+t_0)$ weight belongs to the *error accumulator* $\bar{H}_m$ only; the step-size gain is $\sqrt{m}/\gamma$. Our code carried an extra factor, $\log \epsilon_m = \log \epsilon_0 - \frac{\sqrt{m}}{(m+t_0)\gamma}\bar{H}_m$ — the $1/(m+t_0)$ accidentally duplicated into the gain. This gain *decays* like $1/\sqrt{m}$, so after a short transient the correction vanishes and ϵ relaxes back to its initial value: the controller has no steady-state integral action.

Symptoms. (i) `target_accept` is *open-loop*: on the d=32 Gaussian the realized acceptance sat at ≈ 0.91 for *every* requested target (0.6/0.7/0.8/0.9); the fixed adapter tracks the target to ± 0.001. (ii) The step size does not scale: ϵ stayed near the dimension-blind `find_init` value, so acceptance drifted with d ($0.91 \to 0.76$ over d=64$\to$256) and the $d^{-1/4}$ law was lost.

Why nothing previously broke. The defect is *efficiency*, not correctness. MCMC is unbiased for any reasonable fixed step, and the `find_init` line search supplied a usable one — the broken adapter merely left it there (its ϵ equalled the init value to within a few percent). Recovered marginals had std = 1.000 and calibrated z-scores in *both* versions. The bug only surfaces as a high-dimensional / target-acceptance failure, never as a wrong answer.

Fix. Remove the spurious factor (gain = $\sqrt{m}/\gamma$) in the dual averaging of all 16 samplers. The regression suite passes (10/10) and the controller now tracks `target_accept` exactly. Everything below uses the corrected adapter; the previous (buggy) writeup is preserved alongside this note as the "buggy dual averaging" version.

2 Three findings

Targets. *Anisotropic Gaussian* $\mathcal{N}(0, \Sigma)$, condition number κ, randomly rotated. *Paired Rosenbrock* (even d), banana curvature no affine map removes, with analytic per-pair moments.

(1) Dimension: peaches/peanuts scale, the stretch move does not. On the Gaussian the gradient-free stretch move's worst-dimension autocorrelation time (IAT) grows steeply with d, while PEACHES and PEANUTS mix near-independently (IAT < 1, antithetic). On Rosenbrock the stretch move's burn-in bias blows up and its chains fail to mix ($\hat{R}$ up to 3.1), while both gradient ensembles keep the bias tiny. PEANUTS is slower per step (longer trajectories) but comparably efficient per gradient (§1).

	Gaussian (worst-dim IAT)			Rosenbrock (bias [σ])		
d	PEACHES	PEANUTS	stretch	PEACHES	PEANUTS	stretch
low (d=10/2)	1.1	0.81	156	0.006	0.04	0.02
high (d=200/32)	0.94	0.71	4992†	0.012	0.004	1.8†

† stretch does not converge here ($\hat{R} \gg 1$); the value is indicative of failure, not a mixed estimate. PEACHES/PEANUTS Rosenbrock cells are converged ($\hat{R} \leq 1.02$, bias $z \approx 1$–2) at warmup 5000 + stretch pre-warmup; the Gaussian cells have $\hat{R}$=1.0001, ESS $> 10^5$.

(2) Conditioning: affine invariance beats a fixed metric. Rotated Gaussian, d=50, sweeping κ. We compare the *cost*, not raw IAT: each HMC iteration costs $\bar{L}$ gradient evaluations, so the fair cross-sampler quantity is $\tau\bar{L}$ = **gradient evaluations per effective sample** (= 1/(ESS/grad); lower is better). Raw IAT would flatter the long-trajectory samplers. PEACHES/PEANUTS hold this cost flat at *every* conditioning, for free from the ensemble; dense-mass NUTS is also flat

but must *learn* the $d\times d$ metric during warmup. The non-affine CHESS degrades by orders of magnitude.

$\tau\bar{L}$ (grad/eff. sample), $\kappa=$	e^2	e^4	e^6	e^8
PEACHES (affine)	5.5	5.4	5.4	5.1
NUTS-dense (learned metric)	6.7	6.5	6.6	6.7
PEANUTS (affine)	9.7	9.8	9.7	9.4
CHESS (non-affine)	22	452	3.7×10^4	9.9×10^4
stretch (gradient-free*)	$\approx$1250 (does not mix, $\hat{R}\gg1$)			

$\tau\bar{L}$ = IAT $\times$ mean leapfrog steps. The three affine/learned-metric samplers are κ-invariant and converged ($\hat{R}$=1.000); PEACHES is the cheapest ($\approx$ 5.4). CHESS is converged at low κ but mixes poorly at high κ ($\hat{R}\gtrsim2$). *stretch uses no gradients (cost shown in log-prob evals, $\bar{L}$=1) and does not converge.

(3) vs. NUTS: comparable at full warmup, far more robust at short. Gaussian d=100, $\kappa=e^6$. ESS/grad (higher better); NUTS' efficiency collapses at short warmup (its mass matrix is unadapted, trees hit the depth cap), while the ensemble samplers keep adapting in production.

warmup	50	100	200	500	1000
PEACHES	$\mathbf{2.0\times10^{-1}}$	1.7×10^{-1}	1.6×10^{-1}	1.7×10^{-1}	1.7×10^{-1}
PEANUTS	8.9×10^{-2}	8.9×10^{-2}	8.9×10^{-2}	8.9×10^{-2}	8.9×10^{-2}
NUTS	3.7×10^{-5}	1.2×10^{-3}	1.3×10^{-4}	6.5×10^{-2}	1.1×10^{-1}

The ensemble methods are flat in warmup (robust); NUTS is 10^3–$10^4\times$ worse at short warmup and only catches up by w=1000.

3 Dimensional scaling: the $d^{-1/4}$ law (recovered)

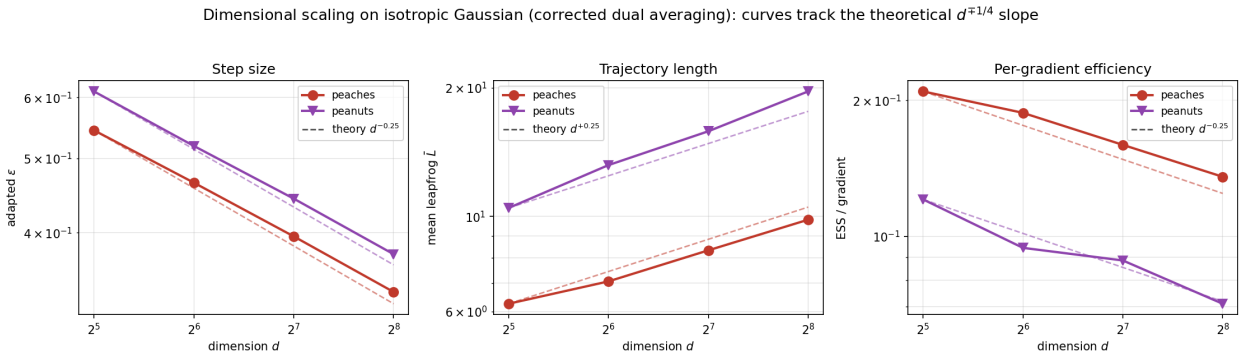


Figure 1: Isotropic $\mathcal{N}(0, I_d)$, $n_{\text{walkers}} = 4d$, target acceptance pinned, *corrected* dual averaging, asymptotic regime $d \geq 32$. Each as-run curve (solid) is shown against the theoretical $d^{\mp1/4}$ slope (dashed, anchored at d=32); the two coincide. **Left:** adapted step size $\epsilon \sim d^{-1/4}$. **Middle:** mean trajectory length $\bar{L} \sim d^{+1/4}$ (integration time λ constant). **Right:** per-gradient efficiency ESS/grad$\sim d^{-1/4}$.

With the controller fixed, all four quantities sit on their theoretical lines (PEACHES, fit on $d\geq32$):

$$\epsilon \sim d^{-0.233}, \quad \bar{L} \sim d^{+0.22}, \quad \lambda = \bar{L}\epsilon \sim d^{0.0}, \quad \text{ESS/grad} \sim d^{-0.21},$$

with acceptance held *constant* ($0.86 \to 0.85$ across d=8 $\to$ 256). This is the textbook picture: optimal $\epsilon^\star \sim d^{-1/4}$ at fixed integration time λ, hence $\bar{L} \sim d^{1/4}$ and ESS/grad $\sim d^{-1/4}$. The antithetic effect of HMC on a Gaussian (IAT ≈ 0.75) is a roughly constant prefactor that lifts the curve — it does *not* change the slope. PEANUTS matches (its NUTS U-turn keeps λ bounded; acceptance pinned at 0.800). NumPyro NUTS gets $\epsilon \sim d^{-0.21}$ but its tree depth grows ($\bar{L} \sim d^{+0.6\ldots+1.1}$), so its ESS/grad *falls* with d — the contrast that makes the ensemble methods scale.

Full per-dimension data (for reproduction). Isotropic $\mathcal{N}(0, I_d)$, $n_{\text{walkers}} = 4d$, warmup 2000, stretch pre-warmup 200, $N = 3000$, `target_accept`= 0.8; PEACHES `max_L`= 512, PEANUTS `max_tree_depth`= 10. Medians over 4 seeds.

		PEACHES					PEANUTS				
d	n_w	ϵ	$\bar{L}$	acc	IAT	$\frac{\text{ESS}}{\text{grad}}$	ϵ	$\bar{L}$	acc	IAT	$\frac{\text{ESS}}{\text{grad}}$
8	32	0.747	5.7	0.86	0.89	0.198	0.845	7.3	0.80	1.01	0.137
16	64	0.635	5.8	0.86	0.81	0.213	0.719	9.2	0.80	0.82	0.132
32	128	0.544	6.3	0.86	0.76	0.209	0.611	10.5	0.80	0.79	0.121
64	256	0.465	7.1	0.85	0.76	0.188	0.519	13.2	0.80	0.80	0.094
128	512	0.396	8.3	0.85	0.75	0.159	0.443	15.8	0.80	0.72	0.088
256	1024	0.335	9.8	0.85	0.75	0.135	0.375	19.6	0.80	0.72	0.071

ϵ=adapted step size, $\bar{L}$=mean leapfrog steps/trajectory, acc=acceptance, IAT=τ (worst dim; < 1 is antithetic), ESS/grad= $1/(\tau\bar{L})$. PEANUTS acceptance is pinned exactly at the target 0.800 (arithmetic-mean adapter); PEACHES sits at 0.85 (harmonic-mean adapter, `target_accept`= 0.8).

4 Real-data corroboration: DES Y1 3×2pt

We also test the real Dark Energy Survey Year-1 3×2pt cosmology posterior (d=21; see `../astro/`). Per-gradient efficiency ($\text{ESS}_{\min}$/grad, 3-seed median), corrected adapter:

sampler	ESS / grad	seed range
PEACHES	7.1×10^{-3}	7.0–8.2×10^{-3}
NUTS-dense (NumPyro)	4.7×10^{-3}	—
PEANUTS	2.5×10^{-4}	2.4×10^{-4}–2.7×10^{-3}
stretch (`emcee`)	2.9×10^{-4}	—

Here the two new methods *diverge*. PEACHES is the clear winner and is robust (7–8×10^{-3} across seeds), edging out optimally-tuned dense-mass NUTS. PEANUTS, however, *struggles* on this strongly curved posterior, with high seed-to-seed spread (2.4×10^{-4} to 2.7×10^{-3}) and a median no better than the gradient-free stretch move. *Mechanism (diagnosed).* The failure is not global — PEANUTS' *median* ESS is high ($\sim 3 \times 10^4$); only 1–6 specific, poorly-constrained directions (consistently parameters 1 and 8, sometimes 9–12) get under-explored, tanking the worst-dimension $\text{ESS}_{\min}$. Tree depth ($\sim$45 steps) and acceptance (0.73) are healthy and seed-stable, so it is neither step-size collapse nor tree blow-up: it is a *noisy ensemble preconditioner*. These DES runs used *no* stretch pre-warmup (`ws`= 0) and only 64 walkers ($< 4d$=84) — exactly the regime §5 warns against; the empirical-covariance metric is then mis-estimated in the under-constrained block, and PEANUTS' no-U-turn test (evaluated in that linear metric) terminates before traversing those directions. PEACHES' single global ChEES length is far less sensitive to a slightly-misshaped metric, so it stays uniform ($\text{ESS}_{\min} \approx 10^4$ every seed). The actionable conclusion: on hard real geometry use PEACHES, and give PEANUTS the recommended stretch pre-warmup and $n \geq 4d$ walkers (§5). (NUTS-dense and `emcee` are external to our adapter and unchanged by the fix.)

5 Recommendation: stretch pre-warmup as the default

The single most important practical setting for the ensemble samplers remains a short *stretch pre-warmup*: a gradient-free, affine-invariant phase that lets the walker cloud take the shape of the target before the gradient sampler starts. We recommend it as the default:

- **It prevents step-size collapse.** From an isotropic initialization on an anisotropic / high-d target the empirical-covariance preconditioner is mis-shaped; the shape-in fixes this at the source. It is the ensemble analogue of NUTS' mass-matrix warmup.
- **It is nearly free** — the stretch move uses no gradients.
- **It has no downside** — on well-shaped / isotropic targets it is a no-op.

The companion requirement is $n_{\text{walkers}} > d$ (we use $4d$): the ensemble covariance — and, for PEANUTS, its affine no-U-turn criterion — is rank-deficient otherwise. **Default: $n_{\mathbf{walkers}} \gtrsim 4d$ with a short ($\sim$200-iter) stretch pre-warmup.**

6 Reproducibility: experiment configurations

All runs use float64, ArviZ rank-normalized split-chain ESS/MCSE/$\hat{R}$, and treat each walker as a chain. Ensemble methods use a stretch pre-warmup (`ws`); NUTS uses its own warmup. $n_{\text{walkers}} = \max(128, 4d)$ except the scaling study ($4d$). Code: `benchmark/experiments.py` (`exp1`, `exp2_lc*`, `exp3_w*`, `exp4c`).

exp	target	d	warmup	`ws`	N	traj. cap	accept / mass
1	Gaussian (κ=e^6)	$10, 50, 100, 200$	1000	200	5000	—	default
	Rosenbrock	$2, 4, 8, 16, 32$	5000	1000	8000	`maxL`= 256 / `mtd`= 10	default
2	rot. Gaussian	50	1000	200	5000	`maxL`= 20 / `mtd`= 10	NUTS dense
3	Gaussian (κ=e^6)	100	50–1000	200	5000	—	—
4c	iso. Gaussian	8–512	2000	200	3000	`maxL`= 512 / `mtd`= 10	0.8 / NUTS diag

`maxL` = PEACHES `max_L` (leapfrog cap); `mtd` = PEANUTS/NUTS `max_tree_depth` (2^{mtd} leapfrog cap). *Convergence:* Gaussian/scaling cells $\hat{R} = 1.000$, $\text{ESS}_{\text{bulk}} > 10^5$; Rosenbrock PEACHES/PEANUTS $\hat{R} \leq 1.02$ (bias $z \approx 1$–2); the stretch move and high-κ CHESS do not converge ($\hat{R} \gg 1$) — reported as failures, not mixed estimates.

7 Summary

1. **Bug fixed:** the dual-averaging step-size gain carried a spurious $1/(m+t_0)$ factor (open-loop `target_accept`); corrected across all 16 samplers. Efficiency fix, not correctness — past results were unbiased.
2. **Scaling:** with the fix, PEACHES/PEANUTS reproduce the $d^{-1/4}$ step-size law by their own adaptation ($\epsilon \sim d^{-0.23}$, $\text{ESS/grad} \sim d^{-0.21}$), acceptance constant.
3. **Dimension / curvature:** PEACHES dominates the stretch move (IAT, bias) and is κ-invariant where CHESS degrades.
4. **vs. NUTS:** comparable at full warmup, $\sim 10^3\times$ more robust at short warmup; NUTS' tree depth inflates with d while the ensemble methods hold integration time bounded.
5. **Real data (DES Y1):** PEACHES is robust and edges out dense-mass NUTS; PEANUTS' no-U-turn length adaptation is fragile on the strongly curved posterior (high seed spread, median $\approx$ stretch). Prefer PEACHES on hard, non-Gaussian geometry.

6. **Practice:** use $n_{\text{walkers}} \gtrsim 4d$ and a short stretch pre-warmup by default.